Energy-Efficient Decision-Making With Belief Structures for Dynamic Path Planning in UAV-Enabled 6G Networks

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***Abstract*—This letter investigates autonomous aerial vehicle (UAV) path planning for sixth generation (6G) wireless networks under uncertainty in user locations and network conditions. Existing approaches often fall short in delivering energy-efficient solutions when making decisions under such uncertainty. We propose a novel decision-making framework that integrates Dempster-Shafer belief structures to model uncertainty and the ordered weighted averaging (OWA) operator to represent the decision-maker's attitude. This decision-making process jointly considers channel capacity and UAV energy consumption for communications and flying. An optimization problem is formulated to maximize energy efficiency subject to network constraints, and it is solved via a low-complexity discrete-time algorithm tailored for dynamic environments. Simulation results demonstrate that the proposed approach outperforms baseline methods in terms of energy efficiency, especially under unreliable user position information, underscoring its suitability for robust UAV-assisted communications in emergency scenarios.**

***Index Terms*—Decision-making, path planning, autonomous aerial vehicles, wireless networks.**

I. Introduction

WIRELESS networks evolving toward sixth generation (6G) increasingly exploit autonomous aerial vehicles as mobile, on-demand base stations (base stations) to deliver rapid coverage and resilient communications in areas where terrestrial infrastructure is unavailable or degraded [1]. UAV-assisted networking promises flexible support for latency-sensitive and energy-constrained applications such as emergency response, crowd management, and internet of things (IoT) deployments [2]. However, achieving these benefits in practice requires trajectory and resource allocation strategies that remain robust under partial observability, intermittent transmissions, and dynamically evolving user behavior.

Most existing approaches to UAV trajectory optimization and resource management either rely on deterministic formulations that assume complete knowledge of ground terminal (GT) positions and channel conditions, a rarity in practical applications [3], or employ data-driven methods such as deep reinforcement learning (DRL), which handle nonstationarity but require extensive training, high computational resources, and careful retraining to remain effective across differing environments [4]. Probabilistic decision frameworks mitigate the need for full observability but still depend on accurate probability distributions of GT locations, which are often unavailable in real-world scenarios [5]. These challenges underscore the need for decision-making mechanisms that can operate under partial information, capture the operator's risk pReferences, and remain computationally efficient.

This letter addresses these gaps by proposing a unified decision-making framework that integrates Dempster-Shafer belief structures with the ordered weighted averaging (OWA) operator to jointly handle uncertainty and risk in UAV trajectory design. Belief structures model subsets of plausible GT positions with associated confidence masses, avoiding reliance on a single and potentially inaccurate probability distribution. Combined with OWA-based aggregation, the framework flexibly balances optimistic and pessimistic decision strategies without requiring computationally intensive learning methods. This decision layer is embedded into an energy-aware trajectory optimization problem, solved via a discrete suboptimal algorithm that preserves computational efficiency while achieving near-optimal energy performance under uncertainty.

In contrast to previous works, the proposed approach departs from conventional paradigms and introduces the following distinctive advancements: (i) unlike deterministic optimization methods, it natively handles imprecise information and partial observability; (ii) unlike DRL-based strategies, it avoids heavy offline training and can adapt online through belief updates; and (iii) unlike purely probabilistic OWA schemes, it does not require full probability distributions, thereby increasing applicability to emergency, sparse-data, or adversarial conditions. Collectively, these features enable energy-efficient path planning for UAVs operating in mobility scenarios, where uncertainty and partial knowledge are dominant concerns.

The main contributions of this letter are:

- We propose a unified decision-making framework for UAV path planning under imprecise GT location information by integrating Dempster-Shafer belief structures with OWA-based risk tuning.
- We formulate an energy-efficient optimization problem under GT mobility uncertainty and network constraints, deriving a discrete suboptimal solver that reduces computational overhead, enabling low-latency operation and potential real-time execution.
- We analyze sensitivity to probability of packet loss and OWA weight selection, showing how these design parameters influence energy consumption.

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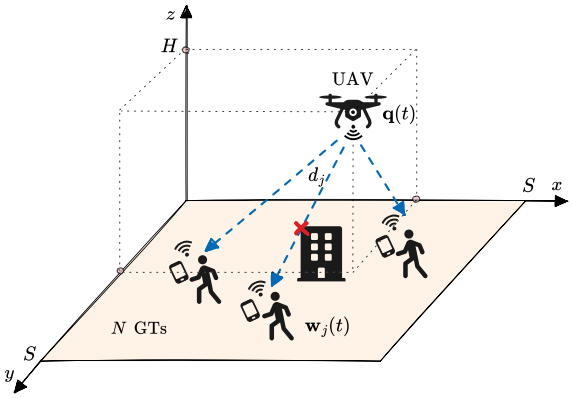


Fig. 1. Wireless communication between a UAV and N GTs.

- We validate the framework through simulations against deterministic, probabilistic, and DRL-based baselines, demonstrating superior energy efficiency and resilience under emergency-like conditions.

II. UAV-Assisted Wireless Network

The wireless network is composed of a UAV acting as a flying BS which moves at a given height H in a two-dimensional (2D) Cartesian coordinate system of side S, and serves N ground terminals randomly located in the area, as shown in Fig 1. The UAV trajectory is described by $\mathbf{q}(t) \in \mathbb{R}^{3\times 1}$ where $0 \leq t \leq T$, and the j-th GT trajectory is denoted by $\mathbf{w}_j(t) \in \mathbb{R}^{3\times 1}$ with zero height. The mobility of both devices makes the Euclidean distance $d_j(t) = \|\mathbf{q}(t) - \mathbf{w}_j(t)\|$ a time-varying parameter that must be minimized.

The communication channel between the UAV and the j-th GT is modeled using a standard and well-established elevation angle-dependent probabilistic line of sight (LOS) framework for aerial-to-ground links [6]. This model captures large-scale fading in urban environments based on statistical building characteristics and incorporates simplified shadowing effects. Accordingly, the path loss is defined under LOS and non-line of sight (NLOS) conditions as follows:

$$PL(t) = \begin{cases} \beta_0 d_j^{-\gamma}(t) & \text{LOS link} \\ k\beta_0 d_j^{-\gamma}(t) & \text{NLOS link,} \end{cases} \tag{1}$$

where β_0 is the free-space path loss at the reference distance $d_0 = 1$ m, the parameter k is the additional attenuation due to NLOS condition, γ is the path loss exponent that typically varies from 2 to 6, and assuming isotropic antennas.

To account for link blockage from obstacles such as buildings, we define the LOS and NLOS probabilities between the UAV and the j-th GT at time t, expressed as functions of the elevation angle $\theta_j(t)$ [6]

$$P_{r_j}^{\text{LOS}}(t) = \frac{1}{1 + a\exp\left(-b[\theta_j(t) - a]\right)}, \tag{2}$$

and

$$P_{r_j}^{\text{NLOS}}(t) = 1 - P_{r_j}^{\text{LOS}}(t), \tag{3}$$

where a and b are related to propagation conditions, and $\theta_j(t) = \frac{180}{\pi}\sin^{-1}\left(\frac{H}{d_j(t)}\right)$ in degrees. Then, the time-varying expected value of the channel coefficient can be derived as:

$$|\overline{h}_j(t)|^2 = P_{r_j}^{\text{LOS}}(t)\beta_0 d_j^{-\gamma}(t) + P_{r_j}^{\text{NLOS}}(t)k\beta_0 d_j^{-\gamma}(t). \tag{4}$$

The downlink channel capacity can be measured to quantify the maximum number of transmitted bits per second (bps) from the UAV for a finite time horizon $T > 0$. One of the main objectives is to control the UAV trajectory to maximize this metric during the simulation period, formally defined as:

$$R_{T_j}(\mathbf{q}(t)) = \int_0^T B\log_2\left(1 + \frac{P_t|\overline{h}_j(t)|^2}{\sigma^2}\right)dt, \tag{5}$$

where B is the bandwidth in Hertz (Hz), P_t is the transmission power by the UAV, and σ^2 is the variance of an additive white gaussian noise (AWGN) random process of zero mean.

Another key objective is to minimize the total energy consumption, as UAVs are battery-powered devices and energy efficiency is critical for sustainable operation. The energy consumption model is divided into two components as follows:

$$E_{T_j}(\mathbf{q}(t)) = E_{c_j} + E_{f_j}(\mathbf{q}(t)), \tag{6}$$

where $E_c = P_c T$ denotes the energy exclusively devoted to data communication, accounting for radiation, signal processing, and circuit operations, with P_c representing the constant communication power. The term $E_f(\mathbf{q}(t))$ corresponds to the propulsion energy required by the mechanical components to keep the UAV airborne. This component dominates the overall UAV power budget and is typically one order of magnitude higher. A detailed energy model for the straight and level flight of a rotary-wing UAV has been generalized in [7] as:

$$\begin{aligned} P_{f_j}(t) &= \underbrace{P_b\left(1 + \frac{3\|\mathbf{v}(t)\|^2}{U_{\text{tip}}^2}\right)}_{\text{blade profile}} + \underbrace{\frac{1}{2}d_0\rho s A\|\mathbf{v}(t)\|^3}_{\text{parasite}} \\ &\quad + \underbrace{P_i\sqrt{1+\frac{a_c^2(t)}{g^2}}\left(\sqrt{1+\frac{a_c^2(t)}{g^2}+\frac{\|\mathbf{v}(t)\|^4}{4v_0^2}} - \frac{\|\mathbf{v}(t)\|^2}{2v_0^2}\right)^{1/2}}_{\text{induced}}, \end{aligned} \tag{7}$$

where P_{f_j} is the propulsion power consumption, with P_b and P_i representing the blade profile and induced power, respectively. In hovering, U_{tip} is the rotor tip speed, v_0 the mean rotor induced velocity, d_0 the fuselage drag ratio, s the rotor solidity, ρ the air density, and A the rotor disc area. The UAV velocity and centrifugal acceleration are given by $\mathbf{v}(t) = \dot{\mathbf{q}}(t)$ and $a_c^2(t)$, respectively, and g denotes gravitational acceleration. The total propulsion energy over the simulation duration T is $E_{f_j}(\mathbf{q}(t)) = \int_0^T P_{f_j}(t)dt + \Delta_K$, where Δ_K accounts for the change in kinetic energy.

On the contrary of the channel capacity, the energy consumption must be minimized at each time slot. Accordingly, we define the energy efficiency as the ratio of the total transmitted bits to total energy consumption:

$$\text{EE}_j(\mathbf{q}(t)) = \frac{R_{T_j}(\mathbf{q}(t))}{E_{T_j}(\mathbf{q}(t))}, \tag{8}$$

which implies that both metrics must be jointly optimized, with propulsion energy typically dominating E_{T_j}.

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III. Proposed UAV Path Planning

The optimal UAV trajectory is computed in real time to ensure energy-efficient network operation by minimizing the distance to ground terminals while improving downlink channel efficiency over time, leading to the following optimization problem:

$$\begin{aligned}\arg\max_{\mathbf{q}(t)}\ & \mathrm{EE}_T = \frac{1}{N}\sum_{j=1}^{N}\mathrm{EE}_j(\mathbf{q}(t)), \forall t \\ \text{s.t. } & \mathrm{EE}_j(\mathbf{q}(t)) \geq \mathrm{EE}_{\min}, \forall j, t \\ & v_{\min} \leq \|\mathbf{v}(t)\| \leq v_{\max}, \forall t \\ & \|a_c(t)\| \leq a_{\max}, \forall t, \end{aligned} \tag{9}$$

where $\mathrm{EE}_{\min}$, $v_{\min}$, $v_{\max}$ and $a_{\max}$ are design parameters.

One approach to identifying a feasible solution involves the consideration of discrete time as $t = nT_s$ where T_s denotes the sample time. This discretization reduces the UAV and ground terminals trajectories to a finite set, denoted as $\mathbf{q}[n]$ and $\mathbf{w}_j[n]$, respectively. Furthermore, we define a finite set of discrete angles that both the UAV and users can take, referred to as $\Theta = \{\theta_1, \theta_2, \ldots, \theta_M\}$, where $\Delta\theta = 2\pi/M$ is the angular resolution. The change in spatial coordinates is then approximated as straight line motion at constant velocity, with abscissa and ordinate projections updated as

$$x[n+1] = x[n] + v\cos(\theta_k)nT_s, \tag{10}$$

$$y[n+1] = y[n] + v\sin(\theta_k)nT_s. \tag{11}$$

A higher level of complexity is introduced by incorporating failures in the user position reports to the UAV, reflecting emergency scenarios with damaged infrastructure and high demand for packet routing through the flying BS. Consequently, the optimization problem is endowed with an element of uncertainty to be managed by the decision maker.

A. Decision-Making With Belief Structures

The decision-making process focuses on selecting the optimal UAV movement at each time slot using Dempster-Shafer belief structures combined with the OWA operator [8]. This approach involves three main components: the set of possible states of nature, corresponding to potential GT positions, denoted as $\mathcal{W} = \{\mathbf{w}_1, \ldots, \mathbf{w}_n\}$; the set of available actions, representing candidate UAV positions, $\mathcal{Q} = \{\mathbf{q}_1, \ldots, \mathbf{q}_m\}$; and the reward, defined as the energy efficiency $\hat{\mathrm{EE}}_{as}$ obtained by choosing action $\mathbf{q}_a$ when state $\mathbf{w}_s$ occurs. The goal is to select the action that maximizes the expected payoff for each UAV-GT link, as represented in the following matrix:

$$\begin{array}{c} \\ \mathbf{q}_1 \\ \vdots \\ \mathbf{q}_a \\ \vdots \\ \mathbf{q}_m \end{array} \begin{array}{c} \begin{array}{ccccc} \mathbf{w}_1 & \cdots & \mathbf{w}_s & \cdots & \mathbf{w}_n \end{array} \\ \begin{bmatrix} \hat{\mathrm{EE}}_{11} & \cdots & \hat{\mathrm{EE}}_{1s} & \cdots & \hat{\mathrm{EE}}_{1n} \\ \vdots & & \vdots & & \vdots \\ \hat{\mathrm{EE}}_{a1} & \cdots & \hat{\mathrm{EE}}_{as} & \cdots & \hat{\mathrm{EE}}_{an} \\ \vdots & & \vdots & & \vdots \\ \hat{\mathrm{EE}}_{m1} & \cdots & \hat{\mathrm{EE}}_{ms} & \cdots & \hat{\mathrm{EE}}_{mn} \end{bmatrix} \end{array}. \tag{12}$$

The procedure assumes that the uncertain position of a GT at each discrete time index is represented by a belief structure m on the states set $\mathcal{W}$. This belief structure consists of a collection of non-empty subsets $\mathcal{U}_k$ known as the focal elements. Each focal element $\mathcal{U}_k$ is assigned a probability mass $m(\mathcal{U}_k)$, satisfying $m(\mathcal{U}_k) > 0$ and $\sum_k m(\mathcal{U}_k) = 1$. Decision-making under this form of uncertainty involves computing a generalized expected value $\hat{\mathrm{EE}}_a$ for each available action (i.e., UAV position) as:

$$\hat{\mathrm{EE}}_a = \sum_{k=1}^{r} V(\mathbf{q}_a, \mathcal{U}_k) m(\mathcal{U}_k), \tag{13}$$

where $V(\mathbf{q}_a, \mathcal{U}_k)$ is the energy efficiency value obtained when the UAV select the position $\mathbf{q}_a$ and the GT position lies in $\mathcal{U}_k$. Then, the UAV must choose the optimal alternative $\mathbf{q}^*$ which provides the largest expected value, that is $\hat{\mathrm{EE}}^* = \max_a \hat{\mathrm{EE}}_a$.

The determination of $V(\mathbf{q}_a, \mathcal{U}_k)$ is performed by means of the OWA operator, which is defined as a weighted average of the elements in the subset $\mathcal{U}_k$ such that [8]:

$$V(\mathbf{q}_a, \mathcal{U}_k) = F(M_{ak}), \tag{14}$$

where M_{ak} denotes the collection of payoffs that can occur under $\mathcal{U}_k$. Each element $\mathbf{w}_s$ in $\mathcal{U}_k$ contributes one element to M_{ak}, hence $M_{ak} = \langle \hat{\mathrm{EE}}_{as} | \mathbf{w}_s \in \mathcal{U}_k \rangle$. The OWA operator, $F : \mathbb{R}^n \to \mathbb{R}$, is a generalization of the weighted average operator, where the weights $\boldsymbol{\omega}$ are not fixed but depend on the rank of the elements in the ordered set. The weights are derived based on the following optimization problem that maximizes the dispersion while guaranteeing a given optimism level β:

$$\begin{aligned}\max_{\boldsymbol{\omega}}\ & \mathrm{Disp}(\boldsymbol{\omega}) = \sum_{i=1}^{c}\omega_i \ln(\omega_i) \\ \text{s.t. } & \mathrm{Opt}(\boldsymbol{\omega}) = \sum_{i=1}^{c}\frac{c-i}{c-1}\omega_i = \beta \\ & \sum_{i=1}^{c}\omega_i = 1, \omega_i \geq 0, \forall j, \end{aligned} \tag{15}$$

where c denotes the cardinality of each focal element, and $0 \leq \beta \leq 1$. Lower values of β reflect a pessimistic stance anticipating adverse GT mobility or propagation conditions, whereas higher values correspond to an optimistic outlook assuming more favorable states. It is worth noting that a larger $\mathrm{Disp}(\boldsymbol{\omega})$ implies a greater influence of the payoffs in computing $F(M_{ak})$.

B. Algorithm Implementation

Algorithm 1 delineates the proposed procedure for determining the optimal UAV trajectory. The algorithm begins by initializing environmental, communication, and energy-related parameters. At each time slot n, the UAV position is updated and payoffs for each candidate action $\mathbf{q}_a$ are evaluated based on the state information of the j-th GT, $\mathbf{w}_{j,s}$, provided by belief structures. The energy efficiency contributions from N ground terminals are then aggregated into a unified payoff matrix. Subsequently, possible states are assigned to each focal element $\mathcal{U}_k$ to determine the corresponding collection of payoffs M_{ak}. The OWA weights, are computed based on the given optimism level β, thereby yielding an intermediate expectation for each focal element. These intermediate expectations are further combined using the probability mass value $m(\mathcal{U}_k)$ to

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Algorithm 1 Decision-Making Based OWA Operator Under Uncertainty (DM-OWA-U)

Input: S, T, N, H, $\mathbf{w}_j[n]$, β, $\mathcal{U}_k, m(\mathcal{U}_k)$
Output: $\mathbf{q}[n]$
for $n \leftarrow 1$ **to** T **do**
  Update $[\mathbf{q}[n]]$
  foreach action $\mathbf{q}_a \leftarrow \mathbf{q}_1$ **to** $\mathbf{q}_m$ **do**
    foreach GT $j \leftarrow 1$ **to** N **do**
      foreach state $\mathbf{w}_s \leftarrow \mathbf{w}_1$ **to** $\mathbf{w}_n$ **do**
        $d_{j,as}[n] = \|\mathbf{q}_a[n] - \mathbf{w}_{j,s}[n]\|$
        Estimate $[h_{j,as}[n] | d_{j,as}[n]]$
        $\hat{\mathrm{EE}}_{j,as} \leftarrow \frac{R_{T_j}(\mathbf{q}_a[n])}{E_{T_j}(\mathbf{q}_a[n])}$
  $\hat{\mathrm{EE}}_{T,as} \leftarrow \frac{1}{N}\sum_{j=1}^{N} \hat{\mathrm{EE}}_{j,sa}$
  forall actions $\mathbf{q}_a$ **do**
    $M_{ak} \leftarrow \mathrm{sort}(\langle \hat{\mathrm{EE}}_{T,as} | \mathbf{w}_s \in \mathcal{U}_k \rangle, \mathrm{descend})$
    Set β and get OWA weights $\boldsymbol{\omega}$
    $V(\mathbf{q}_a[n], \mathcal{U}_k) \leftarrow F(M_{ak})$
  $\hat{\mathrm{EE}}_{T,a} \leftarrow \sum_{k=1}^{r} V(\mathbf{q}_a[n], \mathcal{U}_k) m(\mathcal{U}_k)$
  Optimal reward : $\hat{\mathrm{EE}}_T^* \leftarrow \max_a \hat{\mathrm{EE}}_{T,a}$
  Optimal action : $\mathbf{q}[n]^* \leftarrow \arg\max_a \hat{\mathrm{EE}}_{T,a}$

derive a generalized expectation. Finally, the optimal action is selected as the one that maximizes the expected generalized energy efficiency.

The computational complexity of DM-OWA-U per time slot is $\mathcal{O}\big(m(Nn + rn\log n)\big)$, that linearly scales with the evaluations of UAV actions (m), the number of GTs (N), candidate states or GT positions (n), and the sorting operations required for OWA aggregation over r uncertainty subsets. By comparison, the certainty-based DM-OWA-R [5] runs in $\mathcal{O}(mN)$, and static or circular-trajectory UAV baselines incur $\mathcal{O}(1)$ per evaluation. The increased overhead reflects the adaptive capacity of DM-OWA-U under uncertainty, whereas simpler baselines trade computational cost for reduced flexibility in dynamic environments. The DQN baseline [4] shifts its burden to offline training by requiring intensive batch updates, while maintaining a constant-time forward pass for online inference. In contrast, the proposed approach requires no training and relies on deterministic aggregation, which results in negligible offline cost and lower online complexity.

IV. Multi-User Network Performance

The performance evaluation is conducted using five UAV-enabled schemes: the proposed DM-OWA-U algorithm; the DM-OWA-R algorithm [5] operating under certainty, assuming a probability of one for users being at specific coordinates; a UAV hovering at the center of the field; a circular-trajectory UAV following a path with radius $(\sqrt{2}/2)S$; and a DQN-based UAV with a similar discrete action space [4]; advanced DRL methods are not considered since they incur in even higher complexity than DQN. The DQN agent was trained to maximize energy efficiency under the same environmental conditions for 300 episodes of 2000 steps using ϵ-greedy exploration, experience buffer of 2×10^{-5}, batch size of 128, learning rate of 10^{-3}, and discount factor of 0.99. The Q-network comprises 2 fully connected hidden layers with 256 neurons each and converges after 100 episodes.

TABLE I
Simulation Parameters

Parameter	Value	Parameter	Value
Field length (S)	500 m	Comm. power (P_c)	5 W
GTs / speed (N)	20, 1 m/s	Air density (ρ)	1.225 kg/m^3
UAV height / speed (H, V)	50 m, 10 m/s	Blade power (P_b)	100 W
Time slots (T)	2000	Tip speed (U_{tip})	120 m/s
Elevation parameters (a, b)	10, 0.6	Drag ratio (d_0)	0.3
NLOS factor (k)	0.2	Gravity (g)	9.81 m/s^2
Path-loss exponent (γ)	2.3	Ang. res. ($\Delta\theta$)	45°
Carrier freq. (f_c)	28 GHz	Rotor area (A)	0.5 m^2
Tx power (P_t)	10 dBm	Induced power (P_i)	200 W
Bandwidth (B)	100 MHz	Induced vel. (v_0)	4.03 m/s
Noise power (σ^2)	−110 dBm	Rotor solidity (s)	0.05

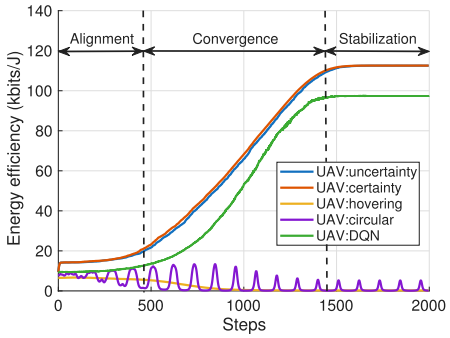


Fig. 2. Energy efficiency for a multi-user scenario with optimism level $\beta = 0.75$ and probability of successful transmission equal to 1.

Autonomous aerial vehicles are initially placed at the center of the field, while ground terminals are randomly distributed. Energy consumption for the circular trajectory is calculated for $a_c = \frac{V^2}{r}$ and $\|v(t)\| = V$ [9]. The remaining parameters are summarized in Table I.

To model unreliable user position reports, uncertainty in GT locations is represented using belief structures with three subsets $\{\mathcal{U}_1, \mathcal{U}_2, \mathcal{U}_3\}$ of potential positions. The probability masses are set to $m(\mathcal{U}_1) = 0.6$ for positions toward the coordinate origin (below the diagonal), $m(\mathcal{U}_2) = 0.3$ for lateral or stationary motion (along the diagonal), and $m(\mathcal{U}_3) = 0.1$ for positions away from the origin (above the diagonal). This configuration reflects documented evacuation behavior in emergencies, where individuals exhibit directional bias toward perceived safe areas rather than random motion [10], with the assigned probability masses encoding the likelihood of movement toward the rescue point. The probability masses encode confidence in different GT movement subsets, shaping UAV decisions and enabling adaptability to varying mobility patterns, while appropriate OWA weight tuning balances optimistic and conservative strategies. For broader deployments, these masses may be estimated or dynamically adjusted by an operator or a sensing layer based on inferred mobility patterns.

Fig. 2 shows the proposed algorithm outperforms the hovering, circular and DQN-based strategies. While DRL methods such as DQN require extensive offline training and do not explicitly capture structured uncertainty or risk preferences, they may be advantageous in fully unknown environments or long-term deployments where training costs can be amortized. The energy efficiency gains are achieved at an optimism

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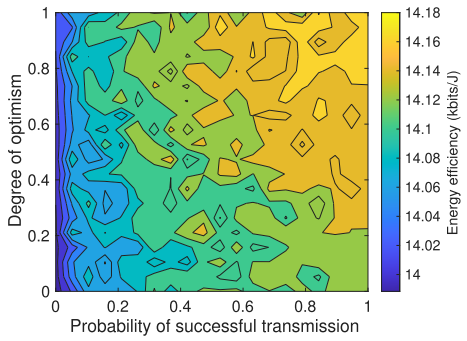


Fig. 3. Impact of optimism level and probability of successful transmission for the first 200 time slots.

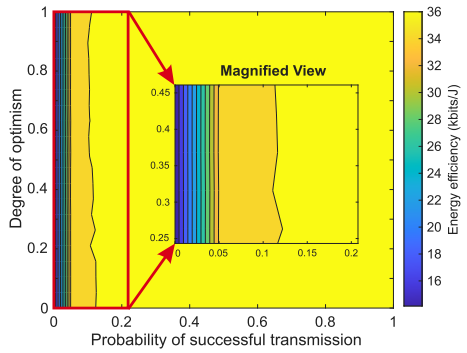


Fig. 4. Average energy efficiency as a function of optimism level and probability of successful transmission over 1400 time slots.

level of $\beta = 0.75$ under perfect transmission of packets containing user position information, and the performance closely matches the certainty case with exact knowledge of GT locations. The results evidence an initial alignment phase where the UAV moves toward the ground terminals with marginal improvements in energy efficiency, followed by convergence phase characterized by steadily increasing efficiency, and final stabilization phase where the UAV hovers to sustain optimal performance.

The impact of the optimism level and the probability of successful transmission on energy efficiency is evaluated by exploring both parameters over the range $[0, 1]$. Since the simulation is carried out as a discrete-time process, a geometric distribution is used to model the number of transmission failures before a success. The probability density function (PDF) is given by: $f(x|p) = p(1-p)^x$, where p is the probability of successful transmission in a single time slot, and x denotes the number of failures preceding the first success.

Fig. 3 presents a contour plot of the average energy efficiency over the first 200 time slots as a function of the optimism level and the probability of successful transmission. As expected, higher optimism levels and mainly probabilities of successful transmission lead to improved energy efficiency, demonstrating the effectiveness of the proposed DM-OWA-U algorithm in adapting to different scenarios. Fig. 4 shows the average energy efficiency for the first 1400 time slots, describing the behavior during the whole convergence phase. The results indicate that the energy efficiency is only affected by low values of probability of successful transmission.

This solution can be deployed in highly dynamic networks, where frequent and abrupt changes in GT positions can increase the computational overhead of belief structure updates. To preserve responsiveness while maintaining tractability, the sampling interval T_s should be carefully chosen, and updates can be selectively triggered for high-mobility ground terminals or significant state variations. Additionally, a more conservative degree of optimism can be applied to the OWA weights to mitigate the effects of high uncertainty. This design balances adaptability and efficiency, enabling real-time feasibility dependent on the system parameters (N, m, n, r) and the underlying hardware for deployment in dynamic UAV-enabled wireless networks.

V. Conclusion

This letter introduces a novel decision-making framework for UAV path planning in multi-user wireless networks under uncertainty. By integrating Dempster-Shafer belief structures with the OWA operator, the proposed DM-OWA-U algorithm effectively models uncertainty in user positions and incorporates the decision-maker's risk attitude. Simulation results confirm that the approach achieves superior energy efficiency compared to baseline strategies, closely approaching the performance of ideal certainty method. The analysis of optimism levels and transmission reliability demonstrates the algorithm's robustness and adaptability. Future work will include a comprehensive sensitivity analysis, adaptive mass assignment for focal elements, and the evaluation of advanced DRL methods to further enhance UAV path planning.

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